MedChain

Use Case Reference Document

Medical Product Passport — NATO-Anchored Use Cases for Total Defence

Internal Consortium Reference | All Work Packages

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Purpose and Scope of This Document

This document defines three reference use cases for the MedChain Medical Product Passport (MPP) platform. It is intended as a shared reference for all Work Package leads, providing a common operational picture that guides technical architecture, resilience modeling, pilot design, standards alignment, and dissemination activities.

Each use case is anchored in NATO's Article 3 baseline requirements for national resilience, specifically, Baseline Requirement 5 (the ability to deal with mass casualties and disruptive health crises) and Baseline Requirement 6 (resilient civil communications systems). This grounding ensures that MedChain's prototype demonstrates capability that is both technically credible and strategically significant for Sweden's Total Defence posture and NATO integration.

The use cases operate at TRL 3–4. Their purpose is to demonstrate that the MPP architecture functions correctly and that the path to a mature system (TRL 5–7) is credible and well-understood. They are not intended to resolve every downstream regulatory or compliance question — those belong to higher TRL phases.

Use-cases

In total we have described three different use-cases:

- UC-1 /Visualization/ Medical Supply Visibility Under Crisis
- UC-2 /Connectivity/ Resilient Operations Under Degraded Connectivity
- UC-3 /MAO Interoperability/ Civil-Military Medical Logistics Coordination.

UC-1 /Visuzalization/
Medical Supply Visibility Under Crisis

NATO Baseline	Baseline Requirement 5 — Ability to deal with mass casualties and disruptive health crises.
Scenario Type	Sudden mass casualty event or rapid-onset health crisis requiring immediate supply chain visibility.
Primary Actors	MSB (Swedish Civil Contingencies Agency), Region Halland, Sahlgrenska University Hospital, AstraZeneca, Tamro, OneMed.
Supporting Partners	Försvarsmakten (situational awareness), SKR (regional coordination), SoftPro (hospital asset systems).
WP Pilot Environment	Region Halland and Sahlgrenska University Hospital (WP5).
TRL Scope	TRL 3–4: Prototype demonstration of real-time stock visibility, product authentication, and redistribution coordination.

Scenario Narrative

A rapid-onset health crisis — triggered by a pandemic surge, a major industrial accident, or a coordinated attack on civilian infrastructure — places immediate and acute demand on Sweden's medical supply chains. Within hours, regional healthcare authorities need to know what critical medical products are available, where they are located, whether they are authentic and serviceable, and how they can be rapidly redistributed to where they are needed most.

Under current conditions, this information does not exist in a single trusted, real-time system. Hospitals, distributors, and authorities each hold fragments of the picture. Manual processes slow decision-making precisely when speed matters most. The MedChain MPP addresses this gap directly.

In this use case, the MPP platform is activated as a crisis visibility tool. Authorized users across the healthcare system — regional administrators, MSB coordinators, and logistics managers at Tamro and OneMed — can query the system to obtain a real-time, verified picture of critical medical stock levels, product locations, and supply chain status.

Key Data Flows

- IoT sensors and digital asset records provide continuous, automated updates on product location, condition, and stock levels across warehouses, distribution hubs, and hospital stores.
- The blockchain layer ensures that all product records are authenticated and tamper-proof — confirming that products are genuine and that their chain of custody is unbroken.
- The MPP dashboard surfaces this data to authorized crisis coordinators in a structured, actionable format — enabling rapid decisions on redistribution, emergency procurement, and priority allocation.

- Digital twin models simulate forward demand scenarios — projecting how long current stocks will last under different crisis intensities and flagging critical shortfall risks before they materialize.

What the Prototype Demonstrates

At TRL 3–4, the prototype demonstrates the following capabilities in a controlled but realistic environment:

Demonstration Goal	Success Indicator	Target (TRL 3–4)
Real-time stock visibility	Authorised users can query verified stock levels across multiple nodes	≥ 3 supply chain nodes connected and queryable in real time
Product authentication	Product authenticity confirmed via blockchain provenance	100% of tracked products verified against blockchain record
Redistribution coordination	System identifies nearest available stock and proposes redistribution routes	Prototype generates redistribution proposal within defined response time
Demand simulation	Digital twin projects stock depletion under crisis scenario	At least 2 crisis scenario simulations validated with partners

Work Package Relevance

Work Package	Relevance to This Use Case
WP1 – Governance	Ensures crisis activation protocols and data access governance are defined; coordinates cross-partner response in pilot scenarios.
WP2 – Platform Architecture	Builds the real-time data ingestion, blockchain verification, and dashboard components that power this use case.
WP3 – Standards & Interoperability	Ensures that stock data from hospitals, distributors, and manufacturers is expressed in interoperable formats aligned with EU DPP and HL7/ FHIR
WP4 – Resilience Modelling	Develops the digital twin demand simulation and supply disruption scenarios; defines crisis activation thresholds.
WP5 – Pilot Testing	Implements and validates UC-1 in Region Halland and Sahlgrenska; collects usability and impact data from MSB and regional coordinators.
WP6 – Dissemination	Documents UC-1 findings as a national reference case; engages MSB and SKR in policy uptake.

UC-2 /Connectivity/
Resilient Operations Under Degraded Connectivity

NATO Baseline	Baseline Requirement 6 — Resilient civil communications systems; ability to maintain critical functions under disrupted connectivity.
Scenario Type	Cyberattack or physical infrastructure disruption severs central network connectivity across one or more regions.
Primary Actors	SkyGrid (edge orchestration), Safespring (sovereign cloud), Region Halland, Försvarsmakten field units.
Supporting Partners	MSB (cyber incident coordination), SoftPro (hospital IT integration), CHORD (risk monitoring).
WP Pilot Environment	Distributed edge nodes across hospital and logistics sites; simulated network degradation (WP4 / WP5).
TRL Scope	TRL 3–4: Prototype demonstration of continued MPP operation under simulated network disruption using edge orchestration.

Scenario Narrative

A sophisticated cyberattack targets Sweden's digital infrastructure, disrupting central cloud connectivity across one or more regions. Alternatively, a physical event — sabotage of a data center, a major power outage, or disrupted telecommunications — produces the same effect. In either case, the central MPP platform loses reliable network connectivity.

This is precisely the moment when supply chain visibility matters most — and when conventional centralized systems fail. Healthcare workers, logistics managers, and defence units in the affected region can no longer access centrally hosted systems. Without a resilient alternative, the crisis response becomes blind.

In this use case, SkyGrid's edge orchestration layer activates automatically. Local edge nodes — pre-positioned at regional hospitals, logistics hubs, and mobile field units — continue operating the MPP in a degraded but functional mode. Critical data is cached locally, decisions can still be made based on the most recent verified state of the supply chain, and the system resynchronizes with the central platform as connectivity is restored.

Key Data Flows

- Edge nodes continuously cache verified product data from the central MPP, maintaining a local state that reflects the most recent confirmed supply chain picture.
- When central connectivity is lost, edge nodes switch to autonomous operation — allowing local queries, authentication checks, and redistribution decisions to continue without central coordination.
- The blockchain layer, operating across distributed nodes, continues to verify product authenticity even in disconnected mode — preventing counterfeit or unverified products from entering the supply chain during a period of vulnerability.
- When connectivity is restored, edge nodes resynchronize with the central platform, reconciling any local decisions or data updates made during the disconnected period.

What the Prototype Demonstrates

At TRL 3–4, the prototype demonstrates continued MPP functionality under simulated network disruption:

Demonstration Goal	Success Indicator	Target (TRL 3–4)
Edge node activation	MPP continues operating at local edge node when central connectivity is severed.	Edge node sustains core functions for defined disruption window.
Local authentication	Blockchain verification operates in distributed mode without central node.	Product authentication confirmed in disconnected state.
Data integrity on resync	Edge data reconciles correctly with central platform on reconnection.	Zero data conflicts or loss on resynchronisation in test scenarios.
Operational continuity	Users at local nodes can query, authenticate, and coordinate during	≥ 95% of core MPP functions available in degraded mode.

Work Package Relevance

Work Package	Relevance to This Use Case
WP1 – Governance	Defines protocols for edge activation, decision authority during disconnection, and resynchronisation governance.
WP2 – Platform Architecture	Designs and builds the edge orchestration layer, local caching architecture, and distributed blockchain validation; integrates SkyGrid components.
WP3 – Standards & Interoperability	Ensures data formats and exchange protocols are consistent between central and edge nodes; defines resynchronisation standards.
WP4 – Resilience Modelling	Designs and executes simulated network disruption scenarios; models degradation thresholds and recovery timelines using digital twin framework.
WP5 – Pilot Testing	Validates UC-2 through controlled disruption tests at pilot sites; collects operational feedback from hospital and logistics users.
WP6 – Dissemination	Documents edge resilience architecture as a transferable model for other critical sectors; contributes to NATO interoperability standards discussions.

UC-3 /Connectivity/
Civil-Military Medical Logistics Coordination

NATO Baseline	Baseline Requirement 5 — Civil support to military operations; continuity of medically relevant services across civilian and defence contexts.
Scenario Type	Crisis or conflict scenario requiring Swedish Armed Forces medical units to rapidly draw on civilian healthcare stocks.
Primary Actors	Försvarsmakten (Swedish Armed Forces), Viable Solutions, Tamro, MSB, Swedish Defence University (FHS).
Supporting Partners	AstraZeneca (pharmaceutical supply), OneMed (distribution), Ministry of Health and Welfare (policy coordination).
WP Pilot Environment	Försvarsmakten logistics test environment; civilian distribution nodes (WP5).
TRL Scope	TRL 3–4: Prototype demonstration of dual-use MPP operation connecting civilian and defence medical supply chains.

Scenario Narrative

Sweden's integration into NATO has made the interoperability of civilian and military supply chains an operational necessity. In a crisis or conflict scenario, the Swedish Armed Forces may need to rapidly draw on civilian medical stocks — pharmaceuticals, medical devices, and consumables held within the civilian healthcare supply chain — to sustain medical readiness for military units in the field.

This requires a trusted, real-time interface between two supply chains that have historically operated in parallel with limited visibility into each other. Försvarsmakten logistics officers need to know what civilian stocks are available, whether they are authentic and serviceable, and how they can be rapidly transferred or allocated to defence units. Civilian healthcare authorities need assurance that any draw-down of civilian stocks is coordinated, proportionate, and does not compromise civilian care capacity.

In this use case, the MedChain MPP serves as the trusted interface between these two worlds. Authorized defense logistics officers gain a structured, read-only view of relevant civilian medical stocks within the MPP. A defined coordination protocol — jointly governed by MSB and Försvarsmakten — allows requests for stock transfers to be initiated, authenticated, and tracked through the same platform that manages civilian supply chain operations.

Key Data Flows

- Civilian MPP data — product locations, stock levels, and authentication records — is made selectively available to authorized Försvarsmakten users through a governed access layer, respecting both GDPR and defense security classifications.
- Defence logistics officers can initiate a stock request through the MPP interface, triggering a coordinated response from civilian distributors and authorities.
- The blockchain layer tracks the full chain of custody as products move from civilian to defence custody — maintaining an authenticated record that can be audited post-crisis.
- Digital twin simulations, developed in WP4, model the impact of military drawdown on civilian supply capacity — allowing authorities to make informed decisions about proportionality and civilian care protection.

What the Prototype Demonstrates

At TRL 3–4, the prototype demonstrates the civil-military interface in a controlled test environment:

Demonstration Goal	Success Indicator	Target (TRL 3–4)
Governed access layer	Defence users access civilian MPP data within defined scope and authorisation.	Role-based access demonstrated with Försvarsmakten test users.
Stock request coordination	A stock transfer request is initiated, tracked, and confirmed through the MPP.	End-to-end request workflow completed in test environment.
Chain of custody tracking	Authenticated record maintained as products move from civilian to defence custody.	Blockchain record verified at each handover point.
Civilian capacity modelling	Digital twin models impact of military draw-down on civilian supply.	At least 1 draw-down scenario simulated with impact assessment.

Work Package Relevance

Work Package	Relevance to This Use Case

WP1 – Governance	Leads design of the civil-military coordination protocol; ensures alignment between MSB, Försvarsmakten, and Ministry of Health and Welfare.
WP2 – Platform Architecture	Builds the governed access layer that enables selective, role-based data sharing between civilian and defence users; ensures defence-grade security requirements are met.
WP3 – Standards & Interoperability	Aligns MPP data formats with NATO interoperability requirements; ensures the civil-military interface meets both EU DPP and NATO standards.
WP4 – Resilience Modelling	Develops civil-military draw-down scenario library; models civilian capacity impacts and coordinates multi-agent decision support for allocation decisions.
WP5 – Pilot Testing	Validates UC-3 with Försvarsmakten in their logistics test environment; collects feedback from both civilian and military users; led by Viable Solutions.
WP6 – Dissemination	Documents UC-3 as Sweden's model for dual-use digital value chain infrastructure; contributes findings to NATO standardisation and EU resilience policy forums.

Cross-Cutting Guidance for All Work Packages

The three use cases form an integrated set. They share a common architecture, data standards, and governance model. The following principles apply across all WPs when working with these use cases:

Scope Boundary

MedChain's responsibility is the communication layer — how lifecycle data is structured, exchanged, and made interoperable across actors. The content layer — the substantive medical, regulatory, and compliance requirements that govern which data must exist — is governed by existing regulations and the stakeholders who must comply with them. At TRL 3–4, the task is to demonstrate that the architecture works and that the communication standards can be implemented. It is not to resolve every downstream compliance question.

NATO Alignment

All three use cases are explicitly designed to demonstrate capabilities relevant to NATO's Article 3 baseline requirements. WP leads should reference this alignment in their deliverables, particularly WP3 (standards), WP4 (resilience scenarios), and WP6 (policy uptake). This framing strengthens the strategic narrative for both Vinnova reporting and future EU/NATO dissemination.

Dual-Use Logic

Each use case operates in both civilian and defence contexts. The same MPP architecture, data standards, and platform components serve both. WP leads should design their components with this dual-use requirement explicitly in mind — avoiding design choices that optimize for one context at the expense of the other.

TRL Discipline

These use cases are scoped for TRL 3–4. The deliverable is a validated prototype and a report describing what is needed to move to a mature system. Partners should resist scope creep toward production readiness during this project phase. Questions about full regulatory compliance, production-scale deployment, or commercial rollout belong to the post-project TRL 5–7 pathway and should be documented as future requirements rather than current deliverables.

Document Information

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